

Concept 3 | Properties and Graphs of Trigonometric Functions

English Student Edition • Focused formulas and guided practice

Formula opener

Amplitude

$$y = a \sin(bx + c) + d \Rightarrow |a|$$

Period

$$T = \frac{2\pi}{|b|}$$

Range

$$d - |a| \leq y \leq d + |a|$$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q001 Written

(1) Express $\sin \frac{9\pi}{5}$ as a trigonometric function of an acute angle. (2) Simplify $\sin \theta + \sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi) + \sin(\theta + \frac{3\pi}{2})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q002 Written

(1) Express $\cos(-\frac{\pi}{6})$ as a trigonometric function of an acute angle. (2) Express $\sin \frac{8\pi}{7}$ as a trigonometric function of an acute angle. (3) Simplify $\cos(\frac{\pi}{2} - \theta) \sin(\pi - \theta) - \sin(\frac{3\pi}{2} + \theta) \cos(\pi - \theta)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q003 Written

Solve all three source demands: (1) express $\cos \frac{17\pi}{5}$, $\cos \frac{5\pi}{7}$, $\sin(-\frac{7\pi}{8})$, $\sin(-\frac{19\pi}{8})$ using acute angles; (2) find $\tan(-\frac{\pi}{6})$ and $\tan(-\frac{\pi}{4})$; (3) simplify (a) $\sin \theta + \cos(\pi + \theta) - \cos(\pi - \theta) + \sin(\pi + \theta)$, (b) $\sin(\theta + \frac{\pi}{2}) \cos(\theta + \pi) + \sin(-\theta) \cos(\frac{\pi}{2} - \theta)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q004 **Written**

Simplify $\sin\left(\frac{3\pi}{2} - \theta\right) \sec(\pi + \theta) - \tan\left(\theta - \frac{\pi}{2}\right) \cot\left(\frac{5\pi}{2} + \theta\right).$

Q005 Written

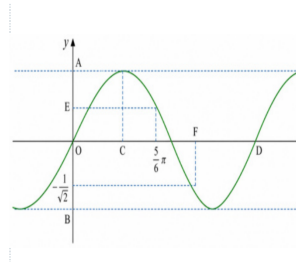
Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, and $y = \tan \theta$, and state each period and range.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q006 Written

For the source graph of $y = \sin \theta$, determine the labelled scales A to F .

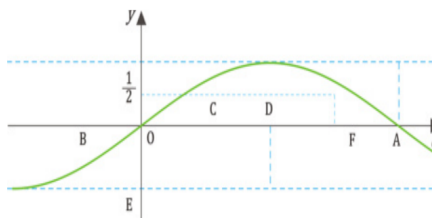


Classified Practice

Properties and Graphs of Trigonometric Functions

Q007 Written

(1) Draw the parent graphs $y = \sin \theta$, $y = \cos \theta$, $y = \tan \theta$, and state their periods. (2) For the source graph of $y = \cos \theta$, determine the labelled scales A to F .

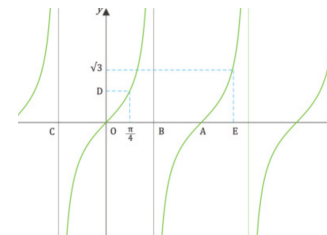
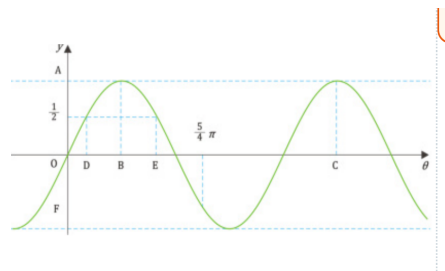
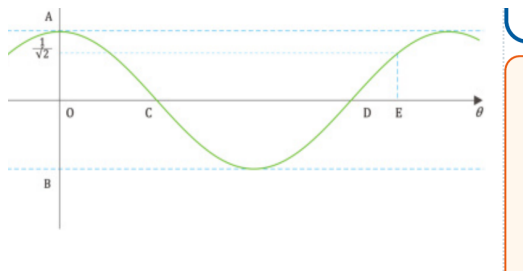


Classified Practice

Properties and Graphs of Trigonometric Functions

Q008 **Written**

Complete all four source graph tasks: draw the three parent graphs and find their periods; read the labelled scales on the source cosine graph; read the labelled scales on the source sine graph; and read the labelled scales on the source tangent graph.

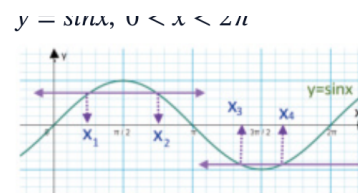


Classified Practice

Properties and Graphs of Trigonometric Functions

Q009 Written

The source graph represents $y = \sin x$ for $0 < x < 2\pi$. Find $x_1 + x_2 + x_3 + x_4$ from the four marked x-coordinates.



Then find: $x_1 + x_2 + x_3 + x_4$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q010 Written

Draw the graphs of $y = 2 \cos \theta$ and $y = \tan \frac{\theta}{2}$, and find their periods.

Q011 Written

Draw the graphs of $y = 2 \sin \theta$, $y = \tan 2\theta$, and $y = \cos \frac{\theta}{2}$, and find their periods.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q012 Written

Draw the nine source graphs and find each period: (a) $3 \sin \theta$, (b) $3 \cos \theta$, (c) $\frac{1}{2} \cos \theta$, (d) $\sin 3\theta$, (e) $\cos 2\theta$, (f) $\tan 3\theta$, (g) $\sin \frac{\theta}{2}$, (h) $\cos \frac{\theta}{3}$, (i) $\tan \frac{\theta}{3}$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q013 Written

Draw $y = -2|\sin(3x)|$ and find its period.

Q014 Written**Draw $y = \sin(\theta - \frac{\pi}{6})$ and find its period.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q015 Written

Draw the graphs and find their periods: (1) $y = \sin(\theta - \frac{\pi}{3})$, (2) $y = \cos(\theta + \frac{\pi}{4})$, (3) $y = \tan(\theta - \frac{\pi}{2})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q016 Written

Draw all source graphs and find their periods: sine group $y = \sin(\theta + \frac{\pi}{4}), \sin(\theta - \frac{\pi}{2}), \sin(\theta + \frac{\pi}{3})$; **cosine group** $y = \cos(\theta - \frac{\pi}{3}), \cos(\theta - \frac{\pi}{4}), \cos(\theta + \frac{\pi}{6})$; **tangent group** $y = \tan(\theta + \frac{\pi}{6}), \tan(\theta - \frac{\pi}{3}), \tan(\theta - \frac{\pi}{4})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q017 Written

Draw $y = 3 \cos \left| x + \frac{\pi}{3} \right|$ and find its period.

Q018 Written

Draw the graphs and find their periods: (1) $y = 2 \cos(2\theta - \frac{\pi}{2})$, (2) $y = \tan(\frac{\theta}{2} - \frac{\pi}{6})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q019 Written

Draw the graphs and find their periods: (a) $y = 3 \sin \frac{\theta}{2}$, (b) $y = \tan(2\theta - \frac{\pi}{2})$, (c) $y = 2 \cos(2\theta - \frac{\pi}{3})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q020 Written

Draw all nine source graphs and find their periods: (a) $2 \cos \frac{\theta}{2}$, (b) $3 \sin 2\theta$, (c) $2 \cos 2\theta$, (d) $2 \cos(\theta - \frac{\pi}{4})$, (e) $2 \sin(\theta - \frac{\pi}{3})$, (f) $\tan(2\theta - \frac{\pi}{3})$, (g) $3 \cos(2\theta - \frac{\pi}{2})$, (h) $3 \sin(2\theta - \frac{2\pi}{3})$, (i) $2 \cos(\frac{\theta}{2} - \frac{\pi}{4})$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q021 Written

Draw $y = 3 \sin |2x - \pi|$ and find its period.

Q022 MCQ $\sin(\theta + \pi)$ equals:**Choose the correct option.****A** $\sin \theta$ **B** $-\sin \theta$ **C** $\cos \theta$ **D** $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q023 MCQ

 $\cos(\theta + \pi)$ equals:

Choose the correct option.

A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q024 MCQ $\sin\left(\frac{\pi}{2} - \theta\right)$ equals:

Choose the correct option.

A $\sin \theta$ B $-\sin \theta$ C $\cos \theta$ D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q025 MCQ

 $\cos(-\theta)$ equals:

Choose the correct option.

A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q026 MCQ **$\sin(-\theta)$ equals:****Choose the correct option.****A** $\sin \theta$ **B** $-\sin \theta$ **C** $\cos \theta$ **D** $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q027 MCQ

 $\cos(\pi - \theta)$ equals:

Choose the correct option.

A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q028 MCQ $\sin\left(\frac{3\pi}{2} + \theta\right)$ equals:

Choose the correct option.

A $\sin \theta$ B $-\sin \theta$ C $\cos \theta$ D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q029 MCQ

 $\tan(\theta - \frac{\pi}{2})$ equals:

Choose the correct option.

A $\tan \theta$ B $-\tan \theta$ C $\cot \theta$ D $-\cot \theta$

Q030 MCQ**The period of $y = \sin \theta$ is:****Choose the correct option.****A** $\pi/2$ **B** π **C** 2π **D** 4π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q031 MCQ

The range of $y = \cos \theta$ is:**Choose the correct option.**

A $y \geq 0$

B $-1 \leq y \leq 1$

C $y \in \mathbb{R}$

D $y > 1$

Q032 MCQ

The period of $y = \tan \theta$ is:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q033 MCQ

A vertical asymptote of $y = \tan \theta$ occurs at:**Choose the correct option.**

A $\theta = k\pi$

B $\theta = \frac{\pi}{2} + k\pi$

C $\theta = 2k\pi$

D $\theta = \frac{\pi}{4} + k\pi$

Q034 MCQ

The maximum value of $\sin \theta$ is:**Choose the correct option.**

A -1

B 0

C 1

D π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q035 MCQ

The x-coordinate of the first positive sine maximum is:**Choose the correct option.**

- A $\pi/4$
- B $\pi/2$
- C π
- D $3\pi/2$

Q036 MCQ**At $\theta = 0$, $\cos \theta$ equals:****Choose the correct option.****A** -1**B** 0**C** 1**D** undefined

Classified Practice

Properties and Graphs of Trigonometric Functions

Q037 MCQ

At $\theta = 0$, $\tan \theta$ equals:**Choose the correct option.**

- A -1
- B 0
- C 1
- D undefined

Q038 MCQ

The range of $y = 3 \sin \theta$ is:**Choose the correct option.**A $[-1,1]$ B $[-3,3]$ C $[0,3]$ D $\mathbb{R}$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q039 MCQ

The period of $y = \sin 4\theta$ is:**Choose the correct option.**A $\pi/4$ B $\pi/2$ C π D 2π

Q040 MCQ**The period of $y = \cos(\theta/3)$ is:****Choose the correct option.**A $2\pi/3$ B 2π C 3π D 6π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q041 MCQ

The period of $y = \tan 5\theta$ is:**Choose the correct option.**A $\pi/5$ B π C 5π D $2\pi/5$

Q042 MCQ

The range of $y = \frac{1}{2} \cos \theta$ is:

Choose the correct option.

A $[-2, 2]$

B $[-1, 1]$

C $[-1/2, 1/2]$

D $[0, 1/2]$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q043 MCQ

The graph of $y = -2 \sin \theta$ is the parent sine graph:**Choose the correct option.**

- A shifted only
- B reflected and stretched vertically
- C compressed horizontally only
- D undefined

Q044 MCQ

The period of $y = \tan(\theta/2)$ is:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q045 MCQ

The period of $y = \sin(3\theta)$ is:**Choose the correct option.**A $\pi/3$ B $2\pi/3$ C 3π D 6π

Q046 MCQ**The graph of $y = \sin(\theta - \pi/4)$ is shifted:****Choose the correct option.**

- A left by $\pi/4$
 - B right by $\pi/4$
 - C up by $\pi/4$
 - D down by $\pi/4$
-
-
-

Classified Practice

Properties and Graphs of Trigonometric Functions

Q047 MCQ

The graph of $y = \cos(\theta + \pi/3)$ is shifted:**Choose the correct option.**A right by $\pi/3$ B left by $\pi/3$ C up by $\pi/3$ D down by $\pi/3$

Q048 MCQ**A horizontal shift changes the sine period from 2π to:****Choose the correct option.****A** π **B** 2π **C** 4π **D** 0

Classified Practice

Properties and Graphs of Trigonometric Functions

Q049 MCQ

A horizontal shift changes the tangent period from π to:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Q050 MCQ**The graph of $y = \sin(\theta + \pi/2)$ equals:****Choose the correct option.**A $\sin \theta$ B $-\sin \theta$ C $\cos \theta$ D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q051 MCQ

The graph of $y = \cos(\theta - \pi)$ equals:**Choose the correct option.**A $\cos \theta$ B $-\cos \theta$ C $\sin \theta$ D $-\sin \theta$

Q052 MCQ

The graph of $y = \tan(\theta - \pi/2)$ has its central zero at:

Choose the correct option.

- A $\theta = 0$
- B $\theta = \pi/2$
- C $\theta = \pi$
- D $\theta = 2\pi$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q053 MCQ

Since cosine is even, $\cos |x|$ equals:**Choose the correct option.**A $\cos(x)$ B $-\cos(x)$ C $\sin(x)$ D $|\cos(x)|$

Q054 MCQ**In $y = a \sin(b\theta - q)$, the horizontal shift is:****Choose the correct option.**

- A q
- B q/b
- C b/q
- D a/b

Classified Practice

Properties and Graphs of Trigonometric Functions

Q055 MCQ

The period of $y = 2 \cos(3\theta - \pi)$ is:**Choose the correct option.**A $\pi/3$ B $2\pi/3$ C π D 2π

Q056 MCQ

The range of $y = 4 \sin(2\theta - 1)$ is:**Choose the correct option.**A $[-1, 1]$ B $[-2, 2]$ C $[-4, 4]$ D $\mathbb{R}$

Classified Practice

Properties and Graphs of Trigonometric Functions

Q057 MCQ

For $y = \tan(2\theta - \pi/3)$, the shift is:**Choose the correct option.**

- A left by $\pi/3$
- B right by $\pi/6$
- C right by $\pi/3$
- D left by $\pi/6$

Q058 MCQ

The period of $y = 3 \sin(\theta/2)$ is:**Choose the correct option.**

- A π
 - B 2π
 - C 3π
 - D 4π
-
-
-

Classified Practice

Properties and Graphs of Trigonometric Functions

Q059 MCQ

The correct transformation order for $y = af(b\theta - q)$ starts with:

Choose the correct option.

- A vertical scale
- B horizontal shift
- C horizontal scale
- D reflection only

Q060 MCQ**The period of $y = \tan(4\theta + q)$ is:****Choose the correct option.****A** $\pi/4$ **B** $\pi/2$ **C** π **D** 2π

Classified Practice

Properties and Graphs of Trigonometric Functions

Q061 MCQ

The range of $y = 3 \sin |2x - \pi|$ is:**Choose the correct option.**A $[-3,3]$ B $[0,3]$ C $[-1,1]$ D $\mathbb{R}$

Q062 Written

Express $\cos \frac{11\pi}{6}$ as a function of an acute angle.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q063 Written

Simplify $\sin(\theta + \pi) - \cos\left(\frac{\pi}{2} - \theta\right)$.

Q064 Written**Simplify** $\cos(\theta + \pi) + \cos(\frac{\pi}{2} - \theta)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q065 Written

Evaluate $\tan\left(-\frac{3\pi}{4}\right)$ **using an acute reference angle.**

Q066 Written

State the period and range of $y = \cos \theta$, then list its key points on one cycle.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q067 Written

State the period and asymptotes of $y = \tan \theta$ on $[-\pi, \pi]$.

Q068 Written

For the parent sine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Q069 Written

For the parent cosine graph, find the x-values where $y = 0$ in $[0, 2\pi]$.

Q070 Written**Find the period and range of $y = 4 \cos \theta$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q071 Written

Find the period and range of $y = \sin(5\theta)$.

Q072 Written**Find the period and range of $y = \tan(\theta/4)$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q073 Written

Describe the transformations from $y = \cos \theta$ to $y = -\frac{1}{2} \cos(2\theta)$.

Q074 Written**Find the shift and period of $y = \sin(\theta + \frac{\pi}{6})$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q075 Written

Find the shift and period of $y = \cos(\theta - \frac{\pi}{2})$.

Q076 Written**Find the central zero and period of $y = \tan(\theta + \frac{\pi}{4})$.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q077 Written

For $y = 2 \sin(4\theta - \pi)$, state the shift, period and range.

Q078 Written**For $y = \cos(\theta/3 + \pi/6)$, state the shift and period.**

Classified Practice

Properties and Graphs of Trigonometric Functions

Q079 Written

For $y = 3 \tan(2\theta + \pi)$, state the shift and period.

Q080 Written

For $y = 2 \cos(\theta/2 - \pi/8)$, state the shift, period and range.

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz MCQ 1 Concept Quiz · MCQ

The period of $y = \sin(2\theta - \frac{\pi}{3})$ is:**Choose the correct option.**A $\pi/2$ B π C 2π D 4π

Quiz MCQ 2 Concept Quiz · MCQ

$\sin(\theta + \pi)$ equals:

Choose the correct option.

A $\sin \theta$

B $-\sin \theta$

C $\cos \theta$

D $-\cos \theta$

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz MCQ 3 [Concept Quiz](#) · [MCQ](#)**The range of $y = 2 \cos \theta$ is:****Choose the correct option.**A $[-1,1]$ B $[-2,2]$ C $[0,2]$ D $\mathbb{R}$

Quiz MCQ 4 Concept Quiz · MCQ

The graph of $y = \cos(\theta - \frac{\pi}{4})$ is shifted:

Choose the correct option.

- A left by $\frac{\pi}{4}$
- B right by $\frac{\pi}{4}$
- C up by $\frac{\pi}{4}$
- D down by $\frac{\pi}{4}$

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 5 Concept Quiz · Written

For $y = 3 \sin(2\theta - \pi)$, state the horizontal shift, period and range.

Quiz WRITTEN 6 Concept Quiz · Written

Simplify $\sin(\theta + \frac{\pi}{2}) + \sin(\theta + \pi)$.

Classified Practice

Properties and Graphs of Trigonometric Functions

Quiz WRITTEN 7 Concept Quiz · Written**Draw one cycle of $y = \tan(\theta - \frac{\pi}{3})$ and state its period and asymptotes.**

Quiz WRITTEN 8 Concept Quiz · Written

For $y = -2|\sin(3x)|$, state its period and range and describe the reflection.

Answer key

Use the question number shown on each page.

- Q001** (1) $-\sin \frac{\pi}{5}$ (2) 0
- Q002** (1) $\cos \frac{\pi}{6}$ (2) $-\sin \frac{\pi}{7}$ (3) $\sin^2 \theta - \cos^2 \theta$
- Q003** (1) $-\cos \frac{2\pi}{5}, -\cos \frac{2\pi}{7}, -\sin \frac{\pi}{8}, -\sin \frac{3\pi}{8}$ (2) $-\frac{1}{\sqrt{3}}, -1$ (3a) 0, (3b) $-\frac{1}{\sqrt{3}}$
- Q004** 0
- Q005** $\sin, \cos$: $T = 2\pi, -1 \leq y \leq 1$; $\tan$: $T = \pi, y \in \mathbb{R}$
- Q006** $A = 1, B = -1, C = \frac{\pi}{2}, D = 2\pi, E = \frac{1}{2}, F = \frac{5\pi}{4}$
- Q007** (1) $T_{\min} = 2\pi, T_{\max} = 2\pi, T_{\min} = \pi$; (2) read the marked extrema, intercepts and quarter-period points from the graph
- Q008** Use the standard cycles: sine/cosine have period 2π and range $[-1, 1]$; tangent has period π , range $\mathbb{R}$, and asymptotes at odd multiples of $\frac{\pi}{2}$. The requested A+F labels are read from the marked ticks and extrema in each supplied graph.
- Q009** Read x_1, x_2, x_3, x_4 from the supplied graph and add the four marked coordinates; the result is the graph-reading total.
- Q010** $y = 2 \cos \theta$: $T = 2\pi, -2 \leq y \leq 2$; $y = \tan \frac{\theta}{2}$: $T = 2\pi$
- Q011** $T = 2\pi, \frac{\pi}{2}, 4\pi$ respectively; ranges $[-2, 2], \mathbb{R}, [-1, 1]$
- Q012** (a) 2π (b) 2π (c) 2π (d) $\frac{2\pi}{3}$ (e) π (f) $\frac{\pi}{3}$ (g) 4π (h) 6π (i) 3π ; ranges follow the outside coefficient
- Q013** $T = \frac{\pi}{3}, 0 \geq y \geq -2$
- Q014** shift right by $\frac{\pi}{6}, T = 2\pi$
- Q015** Shifts: right $\frac{\pi}{3}$, left $\frac{\pi}{4}$, right $\frac{\pi}{2}$; periods $2\pi, 2\pi, \pi$.
- Q016** All sine/cosine periods are 2π ; all tangent periods are π . The shifts are read directly from the brackets.
- Q017** $y = 3 \cos(x + \frac{\pi}{3})$ since cosine is even; $T = 2\pi$
- Q018** (1) $y = 2 \cos 2(\theta - \frac{\pi}{4}), T = \pi, -2 \leq y \leq 2$; (2) $y = \tan \frac{1}{2}(\theta - \frac{\pi}{3}), T = 2\pi$
- Q019** (a) $T = 4\pi, -3 \leq y \leq 3$; (b) $T = \frac{\pi}{2}$; (c) $T = \pi, -2 \leq y \leq 2$
- Q020** Periods: (a) 4π ; (b) π ; (c) π ; (d) 2π ; (e) 2π ; (f) $\frac{\pi}{2}$; (g) π ; (h) π ; (i) 4π . Ranges are set by the outside factors.
- Q021** $y = 3|\sin(2x - \pi)|, T = \frac{\pi}{2}, 0 \leq y \leq 3$
- Q022** B
- Q023** B
- Q024** C
- Q025** A
- Q026** B
- Q027** B
- Q028** D
- Q029** D
- Q030** C
- Q031** B
- Q032** B
- Q033** B
- Q034** C
- Q035** B
- Q036** C
- Q037** B
- Q038** B
- Q039** B
- Q040** D
- Q041** A
- Q042** C
- Q043** B
- Q044** C
- Q045** B
- Q046** B
- Q047** B
- Q048** B
- Q049** B
- Q050** C
- Q051** B
- Q052** B
- Q053** A
- Q054** B
- Q055** B
- Q056** C
- Q057** B
- Q058** D
- Q059** B
- Q060** A
- Q061** B
- Q062** $\cos \frac{\pi}{6}$
- Q063** $-\sin \theta - \sin \theta = -2 \sin \theta$
- Q064** $-\cos \theta + \sin \theta$
- Q065** 1
- Q066** $T = 2\pi, -1 \leq y \leq 1$; (0, 1), $(\frac{\pi}{2}, 0), (\pi, -1), (\frac{3\pi}{2}, 0), (2\pi, 1)$
- Q067** $T = \pi; \theta = \pm \frac{\pi}{2}$
- Q068** 0, $\pi, 2\pi$
- Q069** $\frac{\pi}{2}, \frac{3\pi}{2}$
- Q070** $T = 2\pi, -4 \leq y \leq 4$
- Q071** $T = \frac{2\pi}{5}, -1 \leq y \leq 1$
- Q072** $T = 4\pi, y \in \mathbb{R}$
- Q073** horizontal compression by 1/2, reflection in x-axis and vertical scale 1/2; $T = \pi$
- Q074** left by $\frac{\pi}{6}, T = 2\pi$
- Q075** right by $\frac{\pi}{2}, T = 2\pi$
- Q076** $\theta = -\frac{\pi}{4}; T = \pi$
- Q077** right by $\frac{\pi}{4}, T = \frac{\pi}{2}, -2 \leq y \leq 2$
- Q078** left by $\frac{\pi}{2}, T = 6\pi$
- Q079** left by $\frac{\pi}{2}, T = \frac{\pi}{2}$
- Q080** right by $\frac{\pi}{4}, T = 4\pi, -2 \leq y \leq 2$
- Quiz MCQ 1** B
- Quiz MCQ 2** B
- Quiz MCQ 3** B

Quiz MCQ 4 B**Quiz WRITTEN 6** $\cos \theta - \sin \theta$ **Quiz WRITTEN 7** $T = \pi; \theta = \frac{5\pi}{6} + k\pi$ **Quiz WRITTEN 8** $T = \frac{\pi}{3}, -2 \leq y \leq 0$; reflected below the**Quiz WRITTEN 5** right by $\frac{\pi}{2}$, $T = \pi, -3 \leq y \leq 3$